EFY TESTED Sani Theo

Make Your Own Quadcopter Drone

Ashwini Kumar Sinha

Small drones are being used to deliver parcels, for videography of events such as marriages, and surveillance of enemy territory. You can make a drone at home without spending a fortune. Presented here is a low-cost quadcopter drone using an F450 frame and brushless DC motors. The author's prototype is shown in Fig. 1.

To make the drone, you need to collect the components listed in Table 1. You may also need jumper wires and connectors for the circuit and battery.

Before jumping to construction and assembly of the project, you need to know some basic things about the drone including its propellers, motor rotation, transmitter, and receiver.

Fig. 1: Author's prototype

Components Required — Table 1

Components	Quantity	Description
Quadcopter frame	1	F450
DC motor	4	1000Kv brushless
Flight controller	1	KK2.2
Electronic speed controller (ESC)	4	30A
Radio transmitter and receiver	1	FlySky FS-CT6B
LiPo battery	1	3S 11.2V, 2200mAh
AA battery	8	1.5V alkaline battery
Propeller (CW)	2	25.4cm (10-inch) R1045 for clockwise rotation
Propeller (CCW)	2	25.4cm (10-inch) L1045 for counter-clockwise rotation
Connector for LiPo battery	1	XT60

Propellers

The propellers come with numbers marked on them, starting with either R or L letter and, sometimes, without any letter. If the number starts with R, it means right or clockwise rotation. If the number following R is 1045 (like R1045), the first two digits (10) indicate length of the propeller in inches (10-inch or 25.4cm in this case). The next two digits (45) indicate the pitch (4.5-inch or 11.43cm) the propeller cuts through the air per revolution.

If the number on propeller starts with L, it of course means the propeller is meant for anti-clockwise rotation. In this project, two propellers are for clockwise rotation and two for anti-clockwise.

With increase in pitch, the thrust provided by the propeller also increases. But that requires higher torque from the motor. So, you need to maintain the right balance between motor and propeller configurations.

Motors

In a quadcopter drone, two DC motors are required for clockwise rotation and two for anti-clockwise rotation. A suitable propeller is attached to each of the motor.

For the quadcopter to fly, each

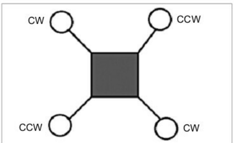

Fig. 2: Quadcopter motors' configuration